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Impact of multidisciplinary medication reconciliation service in older adult patients undergoing thoracic surgery (MERITS) on reducing medication discrepancies: a prospective, controlled, before-and-after study

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Abstract**Background:**

While medication reconciliation (MedRec) is a standard practice in many countries, it is not yet established in Korea. To bridge this gap, we introduced and evaluated a multidisciplinary MedRec service in older adults undergoing thoracic surgery.

Methods:

This single-center prospective study compared the MedRec service (216 patients, September 2022–August 2023) with the usual care group (108 patients, December 2021–April 2022) in older adults undergoing thoracic surgery who were on chronic medication. In the MedRec service, pharmacists obtained the Best Possible Medication History (BPMH) upon admission for use in patient care, unlike usual care, where BPMH was documented post-discharge. Outcomes measured included medication discrepancies at admission and discharge, drug-related problems (DRPs), healthcare visits (emergency department, outpatient clinic and unplanned readmission) within one-month after discharge.

Results:

Medication discrepancies were significantly lower in the intervention group than in the control group at both admission (62.0% vs. 2.8%, $p<0.05$) and discharge (46.3% vs. 0.9%, $p<0.05$). The incidence of DRPs was also lower in the intervention group within the first month post-discharge (32.4% vs. 14.8%, $p<0.05$). However, no significant differences were noted in the rates of emergency department visits (6.5% vs 4.6%, $p=0.66$) or unplanned readmissions (3.7% vs 6.5%, $p=0.44$). In the intervention group, 887 interventions were performed at admission and 88 interventions at discharge, with acceptance rates of 98.8% and 96.6%, respectively.

Conclusion:

The multidisciplinary MedRec service effectively reduced medication discrepancies and DRPs during care transitions. Furthermore, it underscored the importance of pharmacist-led counseling in establishing BPMH and the positive impact of multidisciplinary approach in enhancing medication safety during these transitions.

Trial registration: This study was registered with the Clinical Research Information Service [KCT0006813].

Keywords: transition of care, multidisciplinary, medication reconciliation, medication discrepancy

Background

Care transition is a vulnerable phase for patients, marked by alterations in medication regimens and the potential for medication discrepancies. Communication breakdowns at each stage of the care continuum can result in medication errors and subsequent harm. Systematic reviews have demonstrated that a significant number of patients encounter unintentional medication discrepancies leading to drug-related problems (DRPs) and potential harm [1]. In a previous study, 63.7% of patients experiencing multiple medication changes encountered one or more medication-related problems after discharge [2]. Notably, reconciliation-related errors accounted for 22% at admission, and 12% at discharge, with approximately two-thirds transpiring during care transitions [3]. Earlier research suggests that DRPs contribute substantially to hospital readmissions (40%) [4], with approximately 40% of medication-related hospital readmissions being preventable [5].

To enhance medication safety during care transitions, the Joint Commission introduced medication reconciliation (MedRec service) as a National Patient Safety Goal in 2005, a practice endorsed by international health organizations [6]. The World Health Organization (WHO) has underscored its pivotal role in safeguarding patients from medication-related harm during care transitions [7].

The MedRec service entails compiling an accurate list of an individual's Best Possible Medication History (BPMH) and comparing it with the physician's orders to identify discrepancies and document changes systematically. This process involves patient interviews, interactions with family members or caregivers, and cross-referencing with at least one additional reliable source [6]. Previous systematic reviews consistently demonstrate the significant benefits of the MedRec service in reducing adverse drug events, resulting in a 67% decrease in outpatient visits, a 28% decline in emergency department (ED) visits, and a 19% reduction in hospital readmissions [8]. Furthermore, a systematic review reported a notable 75% reduction in the risk of medication errors when MedRec was implemented [9]. Although MedRec services are globally acknowledged for their role in reducing medication errors [7], their adoption in Korea remains limited due to insurance and structural constraints, particularly during care transitions. Limited information on the prevalence of medication errors in these situations also play a role as a hurdle to broader implementation. While government provides a database service that informs a personal prescribed medication information based on the national insurance system, it operates separately from hospital

electric medical records (EMRs), restricting direct access for healthcare providers. Although some hospitals have initiated MedRec practices, these services have not yet been standardized across the healthcare system.

The Multidisciplinary Medication Reconciliation Service in Adult Patients Undergoing Thoracic Surgery (MERITS) trial aims to address this gap by introducing and evaluating a multidisciplinary MedRec service in Korea. This trial focuses on the impact of MedRec on medication discrepancies during care transitions, highlighting its value for patient safety and care quality during these vulnerable periods.

Methods

Study design

The MERITS study constituted a prospective intervention employing a before-and-after design at a tertiary hospital. The control and intervention groups were enrolled in a 1:2 ratio, with 108 patients in the control group and 216 in the intervention group. The control phase encompassed five months of usual care (December 2021 to April 2022), and the intervention phase extended over 13 months (September 2022 to September 2023) (Figure 1). The study protocol and comprehensive trial methods have been previously published [10] and are accessible in Additional File 1. This study was registered with the Clinical Research Information Service registration number: KCT0006813, Date:07/12/2021.

Setting and study population

The study was conducted in the thoracic surgery wards of a tertiary general hospital. Eligible participants were patients aged 65 years or older who were taking at least one prescribed chronic oral medication. Exclusion criteria included a hospital stay less than 48 hours, inability to comprehend or complete questionnaires, demise during hospitalization, or ineligibility as determined by the medical personnel. Patients were also excluded if they were admitted when a multidisciplinary intervention was not feasible. The intervention group comprised patients who did not participate in the control group. The Clinical Frailty Scale (CFS) score used to assess physical frailty [11].

Control group

In the control group, patients received usual care, with nurses documenting their pre-admission medications. Upon discharge, patients were informed about their medications through a discharge summary prepared by the medical team, without direct involvement from pharmacist. Pharmacists conducted face-to-face consultations with patients at routine physician check-ups two weeks post-discharge. During these consultations, pharmacists retrospectively collected the BPMH at admission and identified medication discrepancies by comparing medications used during the hospital stay with the BPMH, addressing any DRPs identified after discharge. Additionally, telephone follow-ups were conducted within one-month post-discharge to monitor patients' health status and medication-related experiences.

Intervention group

In the intervention group, six clinical pharmacists provided MedRec services, establishing a comprehensive BPMH upon admission from multiple available sources, including nursing records, previous medical records, patient interviews, and government-operated prescription databases. This BPMH was then communicated to physicians, who prescribed medication orders based on this information. Pharmacists resolved discrepancies and, when necessary, implemented additional interventions in collaboration with physicians to address DRPs based on the patient's condition at admission. During the discharge process, pharmacists conducted thorough reviews of discharge medications, corrected discrepancies, provided counseling, and supplied patients with updated medication lists. For weekend discharges, pharmacists conducted telephone follow-ups to ensure continuity of care. Pharmacists also conducted face-to-face consultations two weeks after discharge and a follow-up phone call one month after discharge, as done in the control group.

Outcome measures

Primary outcome was discrepancies between the BPMH and physician orders at admission and discharge. Secondary outcomes included the incidence rate of Medication Appropriate Index (MAI) scores [12], DRPs, healthcare visits (ED, outpatient clinics, and unplanned readmission) within one-month after discharge, and

pharmacist interventions. The classification of DRPs and intervention types followed the guidelines provided by the Pharmaceutical Care Network Europe (PCNE) [13].

Sample size calculation

Based on previous discrepancy prevalence studies [14-17], with a significance level (alpha) of 0.05 and a statistical power of 0.8, accounting for a 10% attrition rate, the study necessitated 216 patients in the intervention group and 108 patients in the control group.

Statistical Analysis

The analysis adhered to intention-to-treat principles, presenting descriptive statistics to summarize baseline characteristics. Categorical variables were assessed using the chi-square or Fisher's exact test, while continuous variables were evaluated using the Student's *t*-test or Mann–Whitney test. Data analysis was conducted using SAS version 9.4 (2017, SAS Institute, Cary, NC, USA), with statistical significance set at $p < 0.05$.

Results

Characteristics of the study population and their preadmission medications

A total of 324 patients participated in the study with 108 in the control and 216 in the intervention group. The mean ages were 71.7 ± 4.7 years in the control group and 72.2 ± 4.7 years in the intervention group. Most patients were classified as "managing well" according to the CFS score. The baseline characteristics of the study participants are shown in Table 1.

Detailed pre-admission medications of the BPMH is shown in Table 2. The intervention group had a significantly higher mean total medication count compared to the control group (8.4 ± 4.3 vs. 6.7 ± 4.1 , $p = 0.001$), with increased use of health supplements and as-needed medications. However, there was no significant difference in the number of prescribed medications between the two groups (control group: 5.8 ± 4.0 ; intervention group: 6.5 ± 4.2).

Impact of MedRec service on medication discrepancies during care transitions

In the control group, medication discrepancies were observed in 62.0% of patients at admission and 46.3% at discharge. In contrast, the MedRec service significantly reduced discrepancies to 2.8% at admission and 0.9% at discharge in the intervention group (Table 3). The primary discrepancy in the control group was omission (59.3%), followed by inaccuracies in dosage, frequency, and administration routes (7.4%). In the intervention group, these discrepancies were reduced, with omissions dropping to 0.9% and inaccuracies in dosage, frequency, or administration routes decreasing to 1.4%. Detailed information is provided in Supplementary Table 1.

Impact of MedRec service on drug-related problems and healthcare utilization post-discharge

The control group demonstrated a higher incidence of DRPs (32.4%) compared to the intervention group (14.8%) ($p < 0.001$). Treatment safety was the predominant DRP type in both cohorts (24.1% in the control group vs. 10.6% in the intervention group), followed by treatment effectiveness (13.0% vs. 5.1%). Drug selection issues were significantly lower in the intervention group compared to the control group (0.9% vs. 22.2%). No significant difference was observed in medication-related health care use between two groups (6.5% in the control group and 4.2% in the intervention group).

Pharmacists' interventions in the intervention group

In the intervention group, pharmacists conducted 887 interventions at admission and 88 at discharge. The majority of interventions at admission were prescriber-level interventions (603), followed by drug-level interventions (284). The most common type of prescriber-level intervention was providing information to the prescriber (316 interventions), particularly related to identifying discrepancies and managing medications that were unavailable in the hospital among the patient's pre-admission medications. The acceptance rates for these interventions were 98.8% at admission and 96.6% at discharge (Table 4). Examples of specific patient cases are provided in Supplementary Table 2.

Discussion

In this prospective study, the MedRec service significantly reduced medication discrepancies during care transitions, decreasing from 62.0% to 2.8% at admission and from 46.3% to 0.9% at discharge. This reduction is consistent with previous research, emphasizing MedRec's critical role in enhancing patient safety during care transitions [14, 18]. Medication omission was the most prevalent discrepancy type, consistent with prior findings [19]. Several factors may contribute to the high discrepancy rate in the control group. At admission, medication interviews primarily focused on oral medications prescribed for chronic conditions, often overlooking as-needed medications, OTC and non-oral medications. Patients also tended not to report health supplements, as they did not perceive them as medications. Additionally, some medications were intentionally omitted by physicians before surgery. However, because the control group review was retrospective, it was not possible to distinguish between intentional and unintentional omissions, resulting in both being classified as discrepancies. At discharge, physicians often prescribe only the medications administered during the hospital stay, frequently without mention of previously prescribed medications from outside the hospital. Both unintentional omissions and undocumented intentional omissions could increase the risk of DRP. With the MedRec service, however, physicians explicitly documented intentional omissions and provided guidance on resuming self-medications, thereby reducing discrepancies and promoting clearer communication. This study underscores the importance of a comprehensive MedRec process to capture all relevant medications and improve communication, ultimately supporting safer transitions in care. Furthermore, significant differences were observed between the control and intervention groups regarding CFS score, and total mean number of medications on the BPMH. Previous studies have suggested that a higher CFS score and a greater number of medications correlate with an increased likelihood of experiencing DRPs [20, 21]. In this study, the intervention group, which comprised a higher proportion of frail patients and a greater number of medications, demonstrated a lower incidence of DRPs compared to the control group. This finding further emphasizes the importance of medication reconciliation in the management of such patients.

The incidence of post-discharge DRPs also significantly decreased from 32.4% in the control group to 14.8% in the intervention group. This reduction was particularly notable in drug selection issues, such as inappropriate drug use or duplication. This enhancement primarily resulted from multifaceted strategies, including comprehensive medication counseling and the provision of a well-documented BPMH upon discharge. Clear elucidation of their current medications ensured that patients were well-informed about their treatments, contributing to a decrease in

DRPs related to drug selection. Also, In the control group, the mean MAI score was 3.4 (± 3.5), which significantly decreased to 0.0 (± 0.2) in the intervention group, indicating an improvement in medication appropriateness due to high intervention acceptance. This reduction aligns with findings from previous studies [22, 23]. Additionally, our study showed very high intervention acceptance rates, with 98.8% at admission and 96.6% at discharge, consistent with previous studies conducted in oncology and neurology setting [24, 25].

There was no statistically significant difference in the number of medication-related healthcare visits within one-month post-discharge between the control and intervention groups. This observation aligns with prior research [26, 27], indicating that while medication reconciliation interventions may improve medication safety, their impact on short-term healthcare utilization may be limited.

Establishing an accurate BPMH was a challenging and demanding process that required integrating information from several sources. To compile BPMH, we relied on interviews with patients or their families, pre-admission medication records completed by nurses, and data from government-operated services, which only provides information on medication within the last year. However, the inconsistencies among these sources pose a challenge in accurately determining the patients' actual medication use.

Our study holds significance as it marks the first prospective controlled examination of a multidisciplinary MedRec service in Korea, systematically conducted to ensure feasibility. Additionally, this investigation yielded valuable insights into the efficacy of medication reconciliation interventions in reducing medication discrepancies and DRPs, underscoring their pivotal role in enhancing patient safety during transitional care. Nevertheless, several limitations were identified in this study. Firstly, the study's single-center approach and specific patient population may limit its generalizability. While a pre-post study design provides valuable insights, it lacks the robustness of a randomized controlled trial, which could have posed ethical challenges for implementation. Furthermore, although the control group phase occurred before the intervention phase, the medical staff could be aware that the study was ongoing. This awareness may have inadvertently influenced their usual care process, leading them to pay closer attention to patients' medications and orders, potentially affecting the outcomes in the control group. Despite employing various sources of information, potential recall bias cannot be entirely ruled out due to the reliance on patient accounts for documenting medication histories in the control group during their

initial post-discharge visit. Secondly, in the process of collecting DRPs at two weeks and the first month post-discharge, it was not always possible to directly verify all DRPs by medical records. We included problems such as medication-related healthcare visits as well as potential problems such as insufficient pain control or mild adverse events based on PCNE classification. Thirdly, this study focused on patients aged 65 and older scheduled for surgery, involving a relatively non-frail cohort. Frailty, often measured using the CFS score, typically denotes a score of 4 or higher [28, 29]; however, only 4% and 13% of patients in the control and intervention groups, respectively, met this criterion. Targeting a higher-risk, frail population, as suggested by previous research [30], might have amplified the impact of medication reconciliation services. Finally, the study coincided with the COVID-19 pandemic, resulting in fewer patients visiting the hospital than usual, which led to longer recruitment times than initially anticipated.

Conclusion

The multidisciplinary MedRec service effectively reduced medication discrepancies and DRPs during care transitions. This study highlights the importance of pharmacist-led patient counseling in establishing BPMH and the positive impact of a multidisciplinary approach in enhancing medication safety during care transitions.

Abbreviations

MedRec: Medication reconciliation

BPMH: Best possible medication history

DRP: Drug related problems

WHO: World Health Organization

ED: Emergency department

MERITS: Multidisciplinary medication reconciliation service in adult patients undergoing thoracic surgery

CFS: Clinical frailty scale

EMR: Electronic medical record

MAI: Medication Appropriateness Index

SD: Standard deviation

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Declarations

Ethics approval and consent to participate

This study was reviewed and approved by the Institutional Review Board of Seoul National University Hospital (IRB No. H-2109-135-1257). All methods were conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all subjects or their legal guardian.

Consent for publication

Not applicable

Availability of data and materials

The dataset used during this study are available from the corresponding author upon reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Author's contribution

All authors contributed to the interpretation of the data and participated in drafting and reviewing the manuscript. HJL served as the principal investigator of the study and obtained grant funding. HJL and J-L contributed to the conception and design of the study. SP (Soyoung Park), AJK, HC, K-H, JYS, SY, SHK, YJJ, JHP, BN, KJN, SP (Samina Park), IKP, CHK, YSC, and YTK were responsible for the implementation of the study. SP(Soyoung Park) and AJK conducted the data analysis. HJL and J-L refined the manuscript with input from all co-authors. All authors critically revised and approved the final manuscript.

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Figure legends

Figure 1. Flow of participants through study

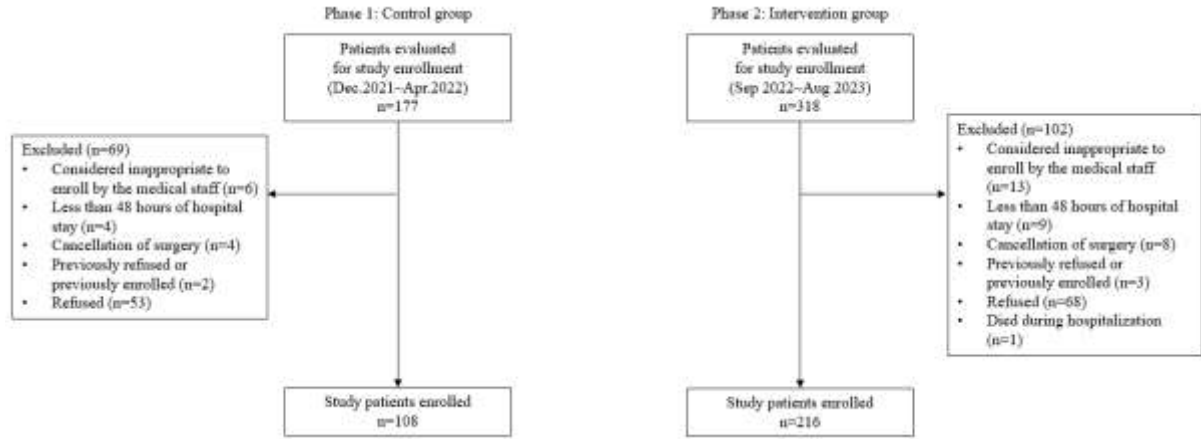


Table 1. Characteristic of study participants

Variables	Control (N=108)	Intervention (N=216)	p-value
	n (%)	n (%)	
Age, mean $\pm$ SD	71.7 $\pm$ 4.7	72.2 $\pm$ 4.7	0.321
Sex (Male)	72 (66.7%)	142 (65.7%)	0.967
Clinical frailty score			0.002
Very fit	3 (2.8%)	0 (0.0%)	
Well	14 (13.0%)	18 (8.3%)	
Managing well	87 (80.6%)	170 (78.7%)	
Vulnerable	3 (2.8%)	28 (13.0%)	
Mildly frail	1 (0.9%)	0 (0.0%)	
Types of thoracic surgery			0.175
Lung	98 (90.7%)	180 (83.3%)	
Esophagus	5 (4.6%)	18 (8.3%)	
Thymus	4 (3.7%)	12 (5.6%)	
Mediastinum	1 (0.9%)	2 (0.9%)	
Pleura	0 (0.0%)	4 (1.9%)	
Total No of medications on BPMH, mean $\pm$ SD	6.7 $\pm$ 4.1	8.4 $\pm$ 4.3	0.001
1~4	36 (33.3%)	37 (17.1%)	
5~9	45 (41.7%)	102 (47.2%)	
More than 10	27 (25.0%)	77 (35.6%)	
Discharge status			0.037
Home	104 (96.3%)	207 (95.8%)	
Hospital	1 (0.9%)	8 (3.7%)	
Nursing home	3 (2.8%)	0 (0.0%)	
Transfer to other wards	0 (0.0%)	1 (0.5%)	
Comorbidity			
Hypertension	67 (62.0%)	130 (60.2%)	0.841
Dyslipidemia	49 (45.4%)	116 (53.7%)	0.195
Diabetes	36 (33.3%)	66 (30.6%)	0.703
BPH	20 (18.5%)	45 (20.8%)	0.731
Coronary artery disease	8 (7.4%)	19 (8.8%)	0.831
COPD	7 (6.5%)	13 (6.0%)	1.000
Osteoporosis	5 (4.6%)	9 (4.2%)	1.000
Arrhythmia	5 (4.6%)	7 (3.2%)	0.755
Hypothyroidism	3 (2.8%)	13 (6.0%)	0.319
Depression	2 (1.9%)	4 (1.9%)	1.000
Length of stay, median (IQR)	6.0 (5.0, 8.5)	6.0 (5.0, 9.5)	0.303
Postop complications	17 (15.7)	49 (22.7)	0.188

SD, Standard deviation; BPMH, Best possible medication history; BPH, Benign prostatic hyperplasia; COPD, Chronic obstructive pulmonary disease; IQR, Interquartile range

Table 2. Characteristics of preadmission medications

Variables	Control (N=108)	Intervention (N=216)	p-value
	n (%)	n (%)	
Total No of medications on BPMH, mean $\pm$ SD	6.7 $\pm$ 4.1	8.4 $\pm$ 4.3	0.001
Prescribed medications, mean $\pm$ SD	5.8 $\pm$ 4.0	6.5 $\pm$ 4.2	0.149
Antihypertensive drugs	67 (62.0%)	138 (63.9%)	0.839
ACE inhibitor, ARBs	48 (44.4%)	94 (43.5%)	0.968
Thiazide	13 (12.0%)	22 (10.2%)	0.752
Beta blockers	11 (10.2%)	28 (13.0%)	0.587
K sparing agents	0 (0.0%)	4 (1.9%)	0.374
Dyslipidemia drugs	64 (59.3%)	143 (66.2%)	0.270
Oral hypoglycemic drugs	33 (30.6%)	67 (31.0%)	1.000
Antiplatelet	27 (25.0%)	59 (27.3%)	0.756
CNS agents	20 (18.5%)	55 (25.5%)	0.209
Benzodiazepines	8 (7.4%)	17 (7.9%)	1.000
Antidepressants	5 (4.6%)	13 (6.0%)	0.797
Antiepileptics	4 (3.7%)	24 (11.1%)	0.043
Z-drug	4 (3.7%)	10 (4.6%)	0.923
TCA	2 (1.9%)	3 (1.4%)	1.000
SSRI, SNRI	1 (0.9%)	6 (2.8%)	0.499
Antipsychotics	0 (0.0%)	3 (1.4%)	0.538
PPI	12 (11.1%)	32 (14.8%)	0.456
Inhaler	5 (4.6%)	17 (7.9%)	0.390
Insulin	2 (1.9%)	7 (3.2%)	0.720
Anticoagulants	4 (3.7%)	8 (3.7%)	1.000
Oral corticosteroids	2 (1.9%)	3 (1.4%)	1.000
NSAIDs	2 (1.9%)	12 (5.6%)	0.209
Health supplements, mean $\pm$ SD	0.7 $\pm$ 1.4	1.1 $\pm$ 1.4	0.026
As-needed medications, mean $\pm$ SD	0.1 $\pm$ 0.4	0.6 $\pm$ 1.3	<0.001
OTC, mean $\pm$ SD	0.1 $\pm$ 0.5	0.1 $\pm$ 0.4	0.600

No, number; BPMH, Best possible medication history; SD, Standard deviation; ACE inhibitors, Angiotensin-converting enzyme; ARBs, Angiotensin receptor blocker; CNS, Central nervous system; TCA, Tricyclic antidepressant; SSRI, Selective serotonin reuptake inhibitor; SNRI, Serotonin norepinephrine reuptake inhibitor; PPI, Proton pump inhibitor; NSAIDs, Nonsteroidal anti-inflammatory drug; OTC, Over the counter drug

Table 3. Impact of medication reconciliation on discrepancies, drug-related problems, and healthcare visits

Variables	Control (N=108)	Intervention (N=216)	P- value
	n (%)	n (%)	
Discrepancy			
at admission	67 (62.0)	6 (2.8)	<0.001
at discharge	50 (46.3)	2 (0.9)	<0.001
MAI score, mean $\pm$ SD	3.4 $\pm$ 3.5	0.0 $\pm$ 0.2	<0.001
Drug related problems within 1 month after discharge	35 (32.4)	32 (14.8)	<0.001
Problems domain			
Treatment safety	26 (24.1)	23 (10.6)	0.003
Treatment effectiveness	14 (13.0)	11 (5.1)	0.022
Others	3 (2.8)	0 (0)	0.070
Cause domain			
Drug selection	24 (22.2)	2 (0.9)	0.001
Patient related	10 (9.3)	11 (5.1)	0.230
Dose selection	8 (7.4)	21 (9.7)	0.630
Treatment duration	1 (0.9)	0 (0)	0.720
All cause healthcare visit within 1 month after discharge	26 (24.1)	63 (29.2)	0.403
ED visit	7 (6.5)	10 (4.6)	0.660
Outpatient visit	21 (19.4)	51 (23.6)	0.479
Readmission	4 (3.7)	14 (6.5)	0.440
Medication-related healthcare visit within 1 month after discharge	7 (6.5)	9 (4.2)	0.526
ED visit	0 (0)	1 (0.5)	1.000
Outpatient visit	7 (6.5)	7 (3.2)	0.245
Readmission	0 (0)	1 (0.5)	1.000
Mortality within 1 month after discharge	0 (0)	0(0)	1.000

MAI: Medication appropriateness index, SD, Standard deviation; ED, Emergency department

Table 4. Pharmacist-led interventions and acceptance of the intervention proposals in the intervention group

Intervention	At admission n (%)	At discharge n (%)
Total	N=887	N=88
At prescriber level	603 (68)	22 (25.0)
Prescriber informed only	316 (35.6)	0 (0)
Prescriber asked for information	3 (0.3)	0 (0)
Intervention proposed to prescriber	284 (32.0)	22 (25.0)
Intervention discussed with prescriber	0 (0)	0 (0)
At patient level	0 (0)	44 (50.0)
Patient (drug) counselling	0 (0)	44 (50)
Written information provided (only)	0 (0)	0 (0)
Intervention proposed to prescriber	0 (0)	0 (0)
Spoken to family member/caregiver	0 (0)	0 (0)
At drug level	284 (32.0)	22 (25.0)
Drug changed to	22 (2.5)	8 (9.1)
Dosage changed to	52 (5.9)	3 (3.4)
Formulation changed to	13 (1.5)	0 (0)
Instructions for use changed to	0 (0)	0 (0)
Drug paused or stopped	21 (2.4)	4 (4.5)
Drug started	176 (19.8)	7 (8.0)
Other intervention or activity	0 (0)	0 (0)
Intervention accepted*	876 (98.8)	85 (96.6)
Intervention accepted and fully implemented	876 (98.8)	83 (94.3)
Intervention accepted, partially implemented	0 (0)	0 (0)
Intervention accepted but not implemented	0 (0)	2 (2.3)
Intervention not accepted	11 (1.3)	1 (1.1)
Intervention not accepted; no agreement	0 (0)	0 (0)
Intervention not accepted; other reason	3 (0.3)	1 (1.1)
Intervention not accepted; unknown reason	8 (0.9)	0 (0)
Other	0 (0)	0 (0)

*Acceptance was also considered in cases where information on prior medication use was reflected in the medical record documentation.